

CosmicWatch Edge Hardware Benchmarks

On-device results: Raspberry Pi 4 vs Jetson Orin Nano — meeting brief, 2026-07-30

Headline results

1. The 49-parameter event classifier runs ~4M events/s on a Pi 4 – about 3,000,000x the real detector rate (1.38 Hz). Inference compute is effectively free on any of these boards.
2. Event-driven gateway on Pi 4: CPU duty fraction at the real event rate is 0.04% – board idle power, not compute, is the entire deployment power budget.
3. FL client (26,584-param autoencoder): Jetson CPU is ~1.8x faster per steady-state round than its own GPU (launch/transfer overhead dominates at this model size), but the GPU trains at barely above idle power and wins ~2.2x on active energy (0.08 vs 0.17 J/round).
4. Jetson idle floor is 3.18 W and software config cannot reduce it (headless vs desktop is a null result; 7W mode saves only ~0.16 W). Low-power deployment needs different hardware, e.g. a microcontroller-class device for the tiny classifier.
5. A Jetson FL client is only justified by GPU-sized models (images / larger GNN); for this workload the Pi 4 at lower idle power remains the right always-on gateway.

Table 1 — FL client local-training benchmark
(real legacy device 13: 2,400 train images, batch 256, 1 local epoch/round)

Board / device	6-round s/round	steady s/round	load W	active W	steady J/round (active)	peak RSS MiB
Pi 4 (4x A72, CPU)	0.900	0.161	not meas.*	—	—	323
Jetson 15W (6x A78AE, CPU)	0.438	0.063	5.95	2.77	0.17	757
Jetson 15W (Orin iGPU, CUDA)	0.481	0.110	3.89	0.71	0.08	1097

* Pi 4 has no software-readable power sensor (vcgencmd reports no current); an inline USB-C meter is required — the result JSON honestly leaves measurement_required set.

Steady-state excludes the one-time ~4 s torch lazy-init cost paid in round 1 (Fig. 3).
Power: INA3221 VDD_IN median, 5 Hz sysfs sampling, taken mid-run during 1000-round (CPU) / 700-round (CUDA) sustained probes.

Table 2 — Tiny event-classifier inference (MLP 4→8→1, 49 params, 49 bytes int8)

Board	numpy us/event	numpy events/s	pure-Python us/event	headroom vs 1.38 Hz detector	active energy per inference
Raspberry Pi 4	75.9	4,218,034	29.9	3,066,100x	not measured*
Raspberry Pi 400	78.2	3,778,277	29.3	2,746,440x	not measured*
Jetson Orin Nano 7W	60.7	5,528,289	24.3	4,018,528x	0.052 uJ
Jetson Orin Nano 15W	38.4	8,815,617	15.3	6,408,096x	0.074 uJ

Pi 400 result carries throttled=0x50000 (past undervoltage) — treat as indicative only.

Federated-learning client benchmark — Pi 4 vs Jetson Orin Nano (15W)

Fig. 1 — Steady-state round time
Jetson CPU ~1.8x faster than its own GPU

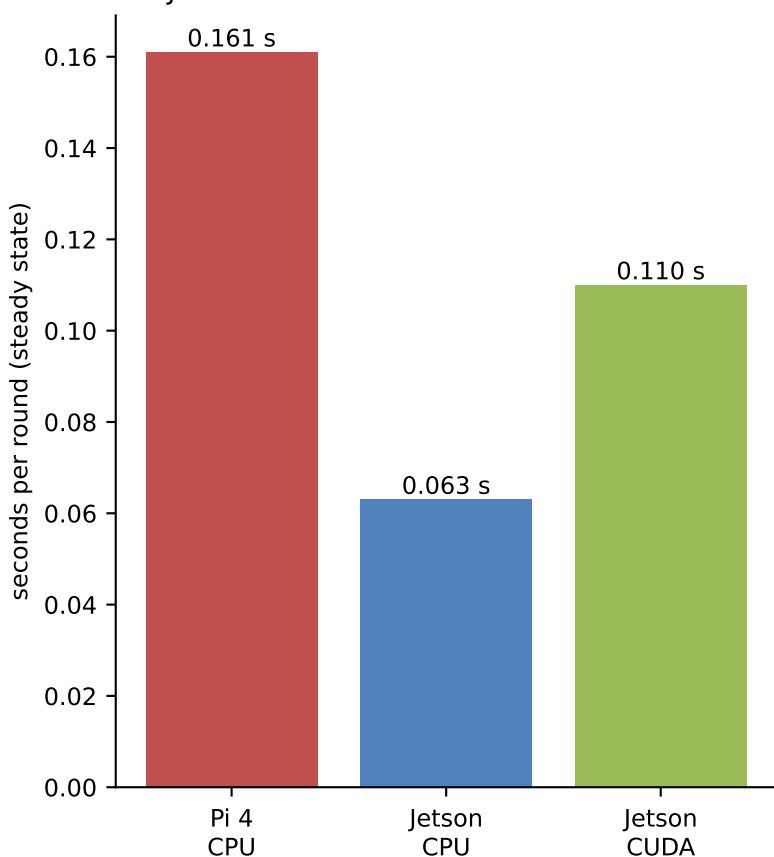


Fig. 2 — Energy per FL round (active)
GPU wins ~2.2x; both are < 0.5 J

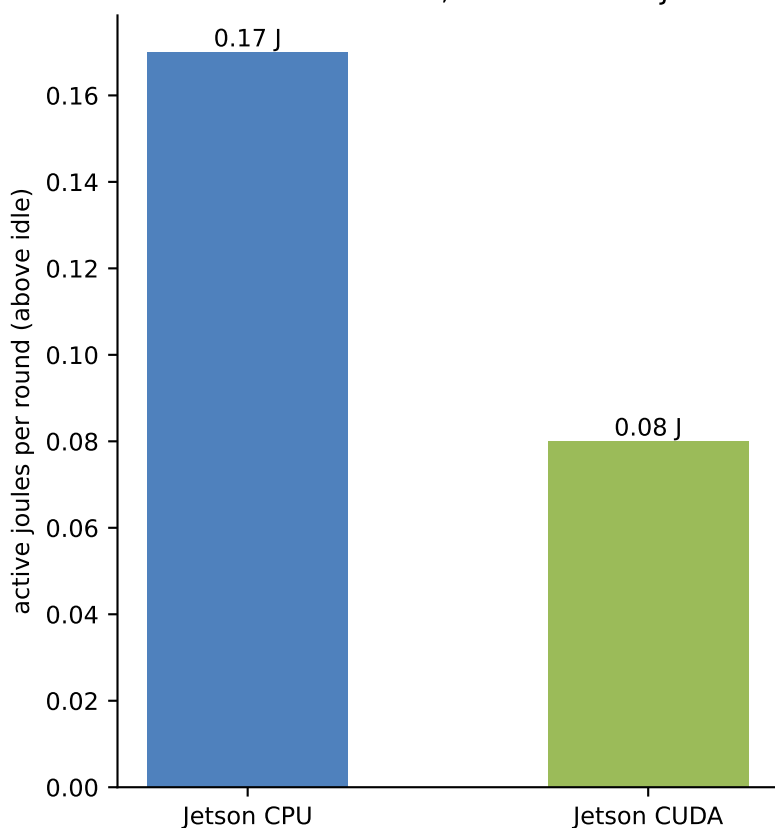


Fig. 3 — Per-round time, 6-round run
round 1 carries ~2-4.6 s one-time torch warmup

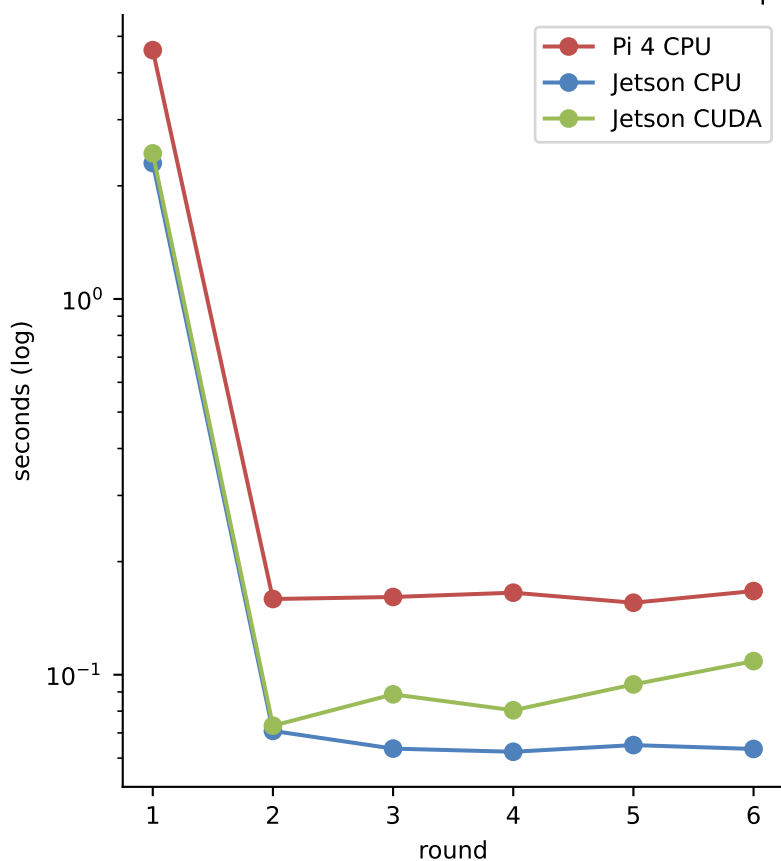
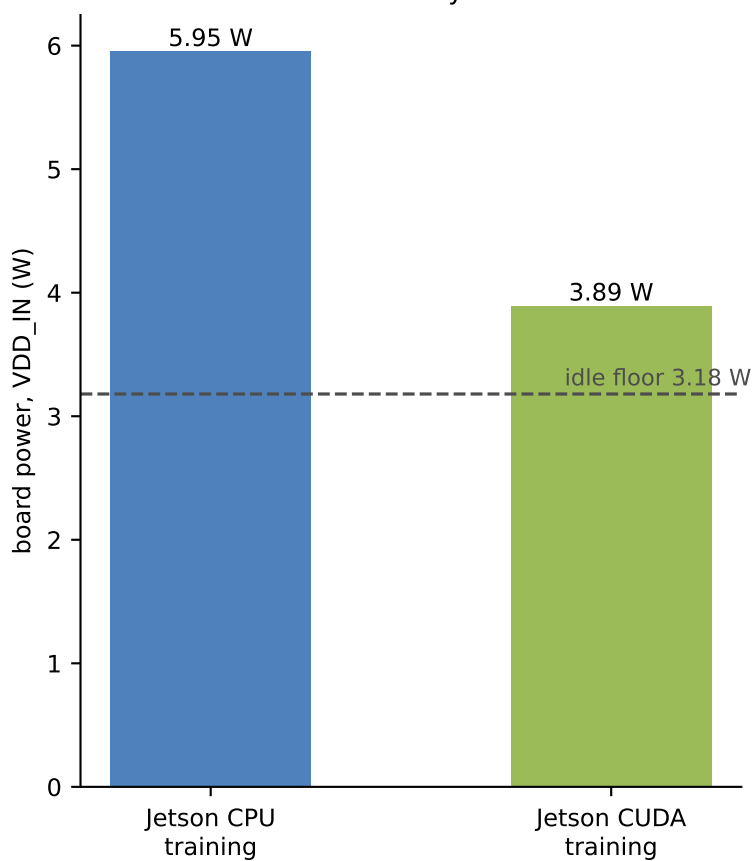


Fig. 4 — Board power under training load
CUDA trains barely above idle



What the FL client trains on — legacy CREDO 20×20 hit-crop images

The benchmarked workload is image data: 65,994 deduplicated CREDO hit-crops split into 8 FL clients by real device ID. The timed client (device 13) trains a 26,584-param autoencoder (400→32→8→32→400, weighted MSE emphasizing bright pixels) on its 2,400 images.

Fig. 9 — 30 random training images from the benchmarked client (device 13, n=2,400)

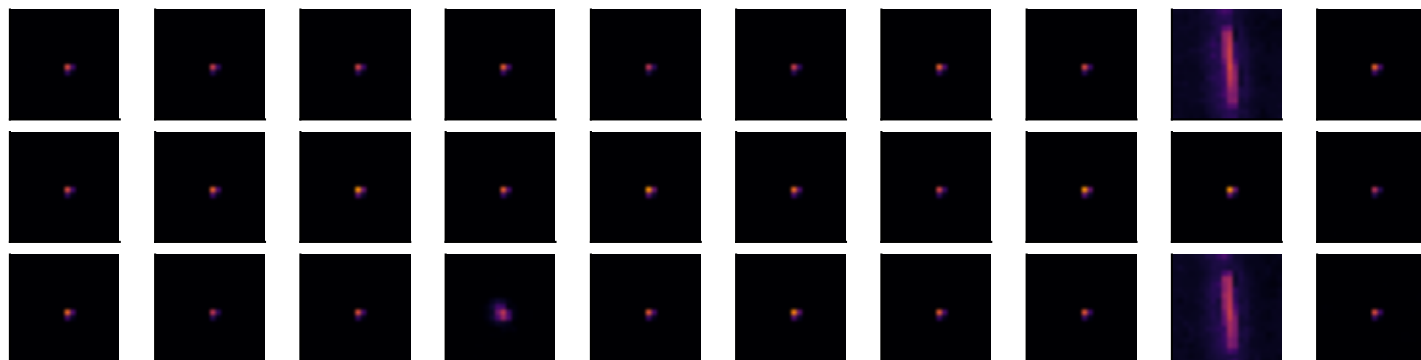


Fig. 10 — The federation is non-IID: 8 real devices, unequal sizes, distinct morphologies (right column: per-client mean image)

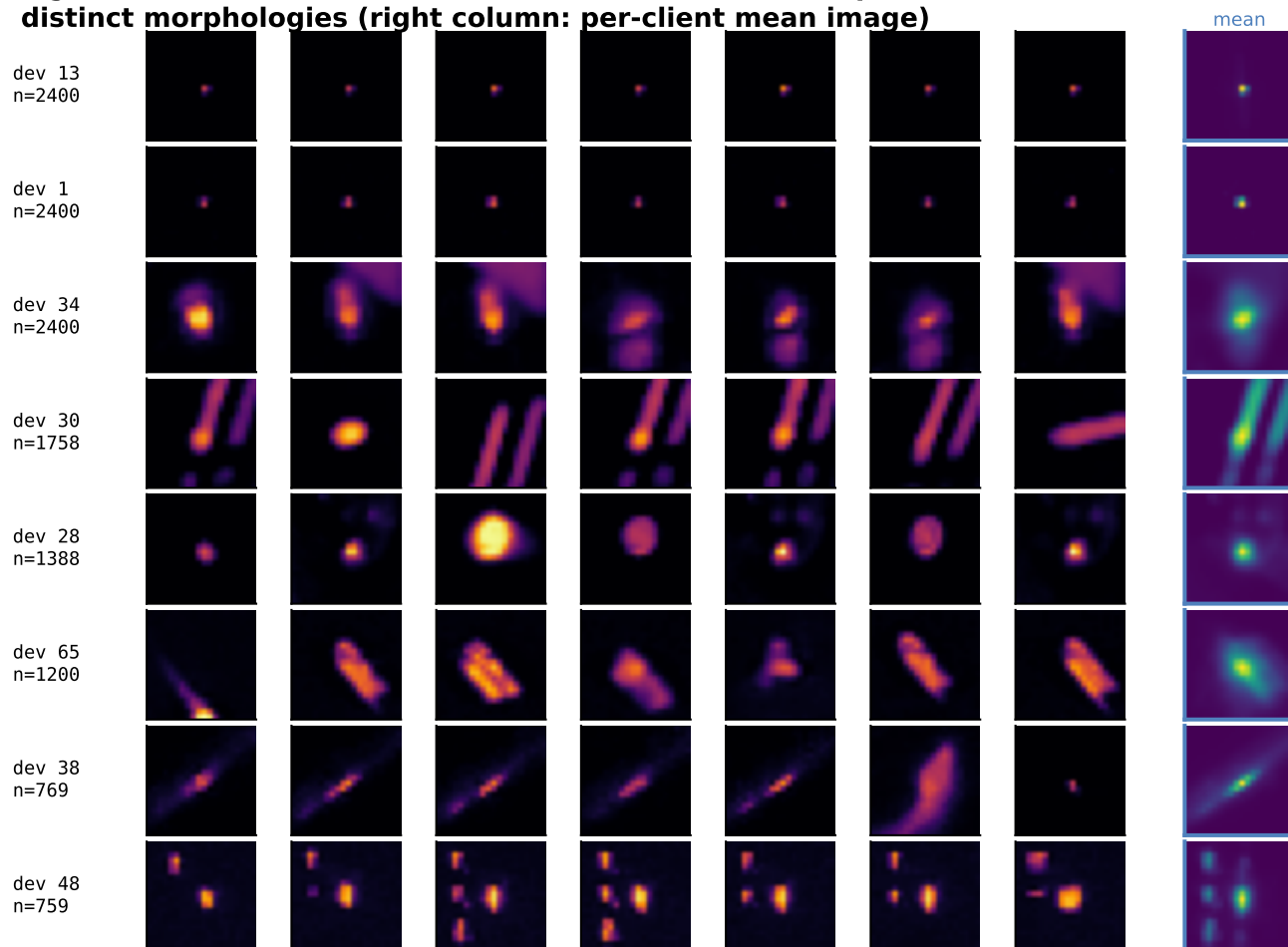
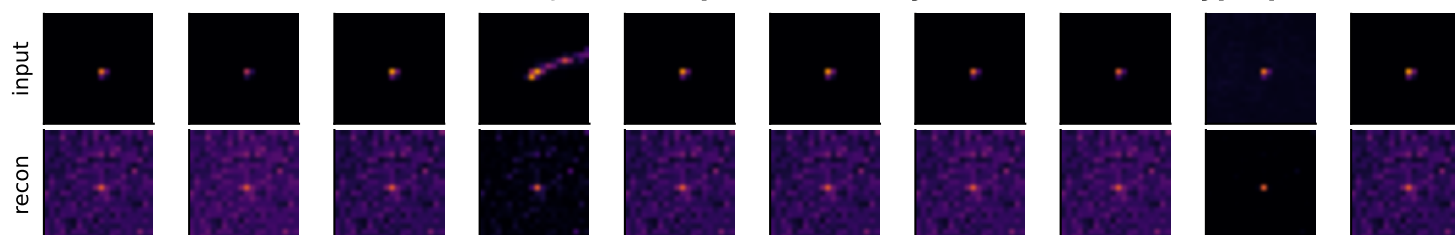


Fig. 11 — Held-out test images (top) vs autoencoder reconstruction (bottom) after the same 6-round run timed on the Pi 4 and Jetson (replicated locally, identical seeds/hyperparameters)



Multi-modal federation demo — one FedAvg layer, two client types

One round scheduler trains and aggregates two separate model heads per round: event clients (tiny MLP, real features from the full 237k-event Jan window, synthetic non-IID partition) and image clients (26,584-param autoencoder, 8 real legacy devices). The federation is the shared layer, not the representation — no cross-modal fusion is claimed (all sources are temporally disjoint).

Fig. 12 — Event head: near-parity federated 0.396 vs centralized 0.407

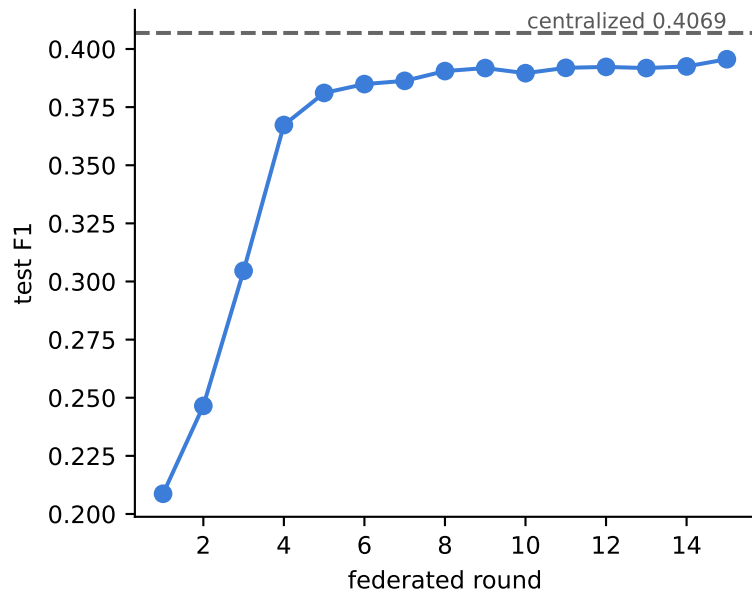


Fig. 13 — Image head: converging across 8 genuinely non-IID devices

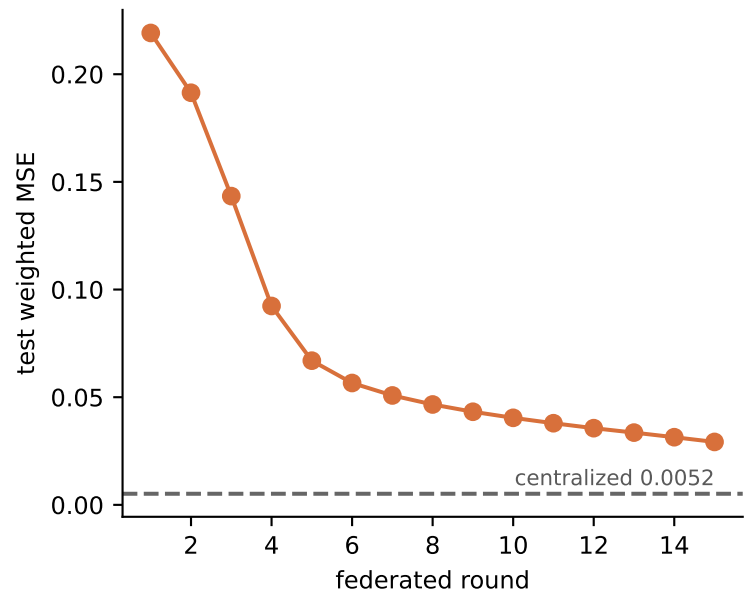
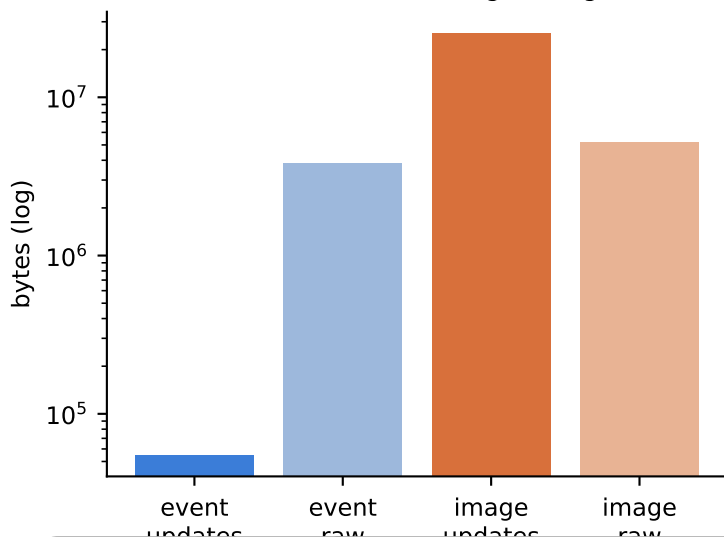


Fig. 14 — Communication is modality-dependent
events: 0.0144x raw (~70x saving) images: 4.88x raw



Conclusions

1. One federation genuinely serves heterogeneous sensor modalities: scheduling and FedAvg aggregation are modality-blind; per-modality model heads are sufficient.
2. The bandwidth argument is modality-dependent. Events: accuracy near-parity at 1.4% of the raw bytes (model << data; the stream grows, the update doesn't). Images: updates cost 4.9x the raw bytes (model >> per-client data) - the image side's case is privacy/decentralization, not bandwidth.
3. Design rule: federate SMALL models over HIGH-RATE streams for bandwidth; federate LARGE models over small private datasets only for privacy/governance. The same infrastructure supports both simultaneously.
4. Non-IID cost is real and visible: the event gap to centralized is small (0.011 F1), but the image gap (0.029 vs 0.005 MSE) is the honest price of never centralizing data from genuinely different devices.
5. Not shown, and why: no shared cross-modal latent space - all sources are temporally disjoint, so per-sample fusion has no data to validate against. The event-side client split remains synthetic (one physical detector).

Inference headroom, gateway duty cycle, and the idle-power story

Fig. 5 — Inference latency (49-param MLP)
all boards $\sim 10^6$ - 10^7 x the detector rate

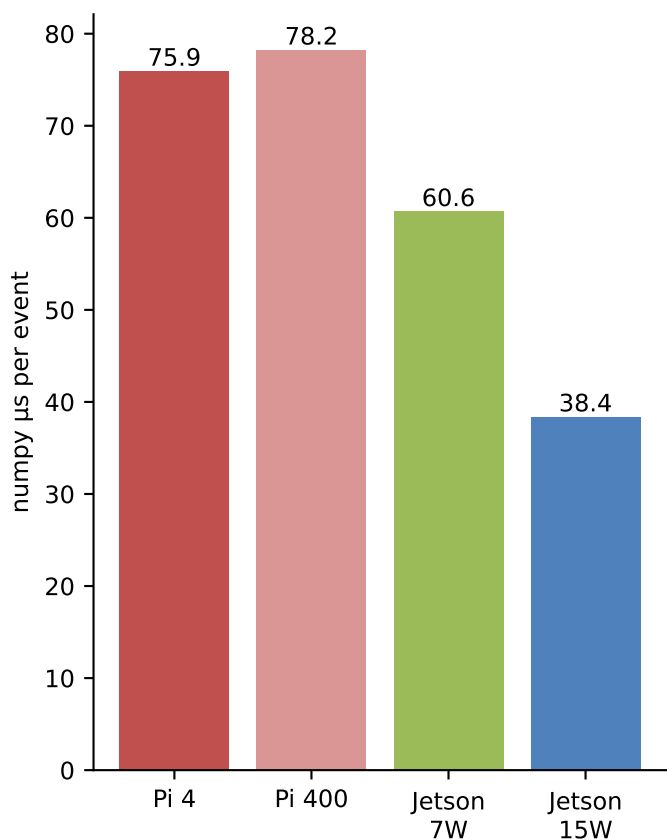


Fig. 6 — Gateway policy CPU cost on Pi 4
hybrid policy: 9,577x headroom, 0.04% duty
at the real 1.38 Hz event rate

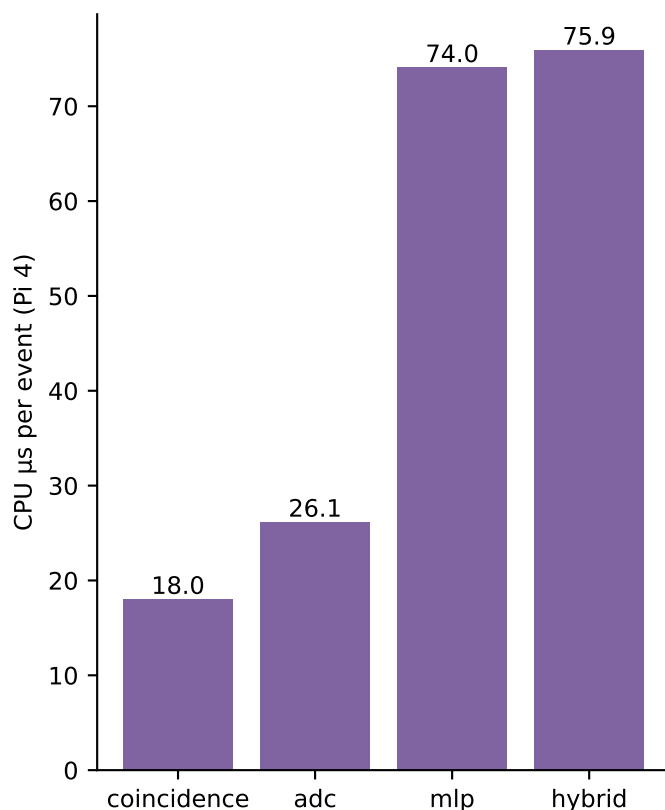


Fig. 7 — Jetson idle power vs software config
headless saves 0.04 W; 7W mode saves 0.16 W

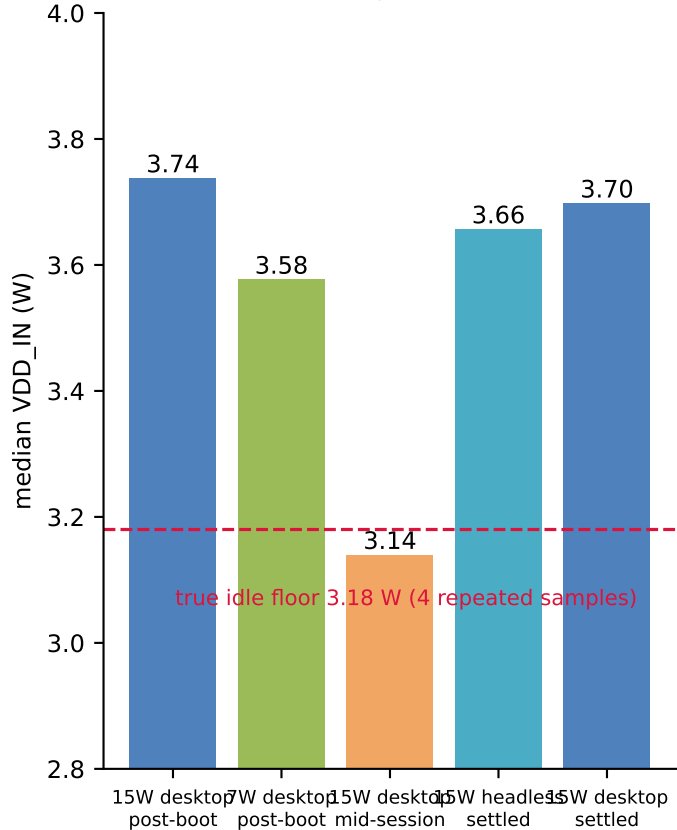
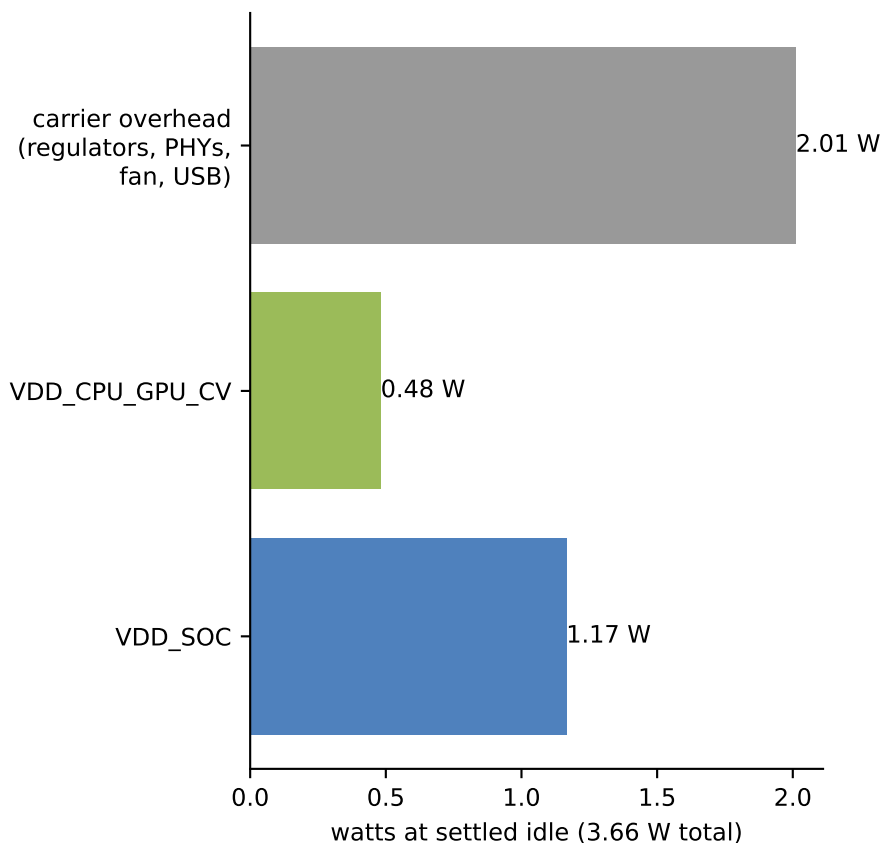


Fig. 8 — Where Jetson idle power goes
most is not software-addressable



Idle-power experiment detail and deployment decisions

Table 3 — Jetson Orin Nano idle-power measurements (INA3221 VDD_IN median, 5 Hz)

Condition	median W	window
15W desktop idle (2026-07-17, post-boot)	3.74	25 s
7W desktop idle (2026-07-17, post-boot)	3.58	25 s
15W desktop idle (2026-07-18, mid-session)	3.14	25 s
15W headless idle (settled, 2026-07-18)	3.66	60 s
15W desktop idle (settled, 2026-07-18)	3.70	60 s

Findings (revised 2026-07-18)

- True deep-idle floor: 3.18 W, rock-stable across four 30 s samples. The 3.66-3.74 W 'settled' readings were an elevated post-activity state (fan hysteresis suspected); idle has two states ~0.5 W apart.
- Desktop vs headless: null result (3.698 vs 3.657 W, ~1%) – GPU rail-gates either way.
- 7W nvpmode saves only ~0.16 W idle while giving up 37% of (unused) throughput.
- Conclusion: software cannot meaningfully reduce the dev-kit idle floor. Low-power deployment levers are hardware: custom carrier, duty-cycling/deep sleep, or moving the 49-parameter classifier to a microcontroller where idle is milliwatts.

Decision criteria (from EVENT_DRIVEN_HARDWARE.md)

- Prefer the Pi 4 as the always-on gateway: lower board power, and compute is irrelevant at a 0.04% duty fraction.
- Prefer the Jetson only when a GPU-sized workload (image training, realistic larger GNN) offsets its higher idle draw – the 26,584-param FL autoencoder does not qualify.
- Communication caveat: the 6-round FedAvg experiment transfers 1.95x the raw grayscale bytes, so compute speed alone cannot make the FL communication case.
- Open measurement: Pi 4 board watts need an inline USB-C meter (options and procedure documented in EVENT_DRIVEN_HARDWARE.md).

Caveats stated honestly

- Gateway selection fractions come from a synthetic stream – hardware cost numbers only.
- JetPack-7 PyPI torch wheel lacks CC 8.7 kernels; CUDA runs via PTX JIT (one-time cost, no steady-state penalty observed).
- 'Event-driven' means kernel-blocking software wakeup, not neuromorphic execution.

CosmicWatch Edge Hardware Benchmarks — Pi 5 Update

Five-device matrix after adding Raspberry Pi 5 (16 GB, Python 3.13, numpy 2.2, torch 2.14 CPU)

Headline results — Raspberry Pi 5 update (2026-09-02)

1. The Pi 5 is now the fastest board in the fleet on ALL THREE harnesses: inference triage, event gateway, and the FL client workload.
2. Tiny-classifier inference: 23.7 us/event single-event, 12.6M events/s batched (3.0x Pi 4, 1.4x Jetson 15W); headroom vs the 1.38 Hz detector rate is ~9.2 million x.
3. Event gateway: 3.8-4.5x the Pi 4 per policy; hybrid CPU duty at the real event rate is 0.016% (Pi 4: 0.041%); headroom 36,215x.
4. FL client: 0.057 s/round steady-state (mean of rounds 2-6, computed uniformly for every device) — faster than the Jetson's own CPU (0.065) and 1.6x its CUDA path (0.089). GPU dispatch still does not pay at this model size; the Pi 5 extends the CPU's win.
5. Energy columns for all Pi rows still await an inline USB-C meter (Jetson rows use onboard INA sensors). Pi 5 note: inline meters may renegotiate 5A PD down to 3A — harmless at these loads.

Caveat: the Pi 5 gateway run uses the committed `cw_gateway_bench.py` driver; the original Pi 4 stream generator was ad hoc, and selection fractions differ (more selected on Pi 5 => timings slightly pessimistic). Only CPU cost, throughput, and duty fraction are hardware measurements.

Table A — Tiny event-classifier inference (49-param MLP, int8 weights)

Board	numpy us/ev	numpy ev/s	pure-py us/ev	headroom vs 1.38 Hz
Raspberry Pi 5	23.7	12,633,241	8.6	9,183,137x
Jetson Orin 15W	38.4	8,815,617	15.3	6,408,096x
Jetson Orin 7W	60.6	5,528,289	24.2	4,018,528x
Raspberry Pi 4	75.9	4,218,034	29.9	3,066,100x
Raspberry Pi 400	78.2	3,778,277	29.3	2,746,440x

Table B — FL client (device 13, 6 rounds, 4 threads)

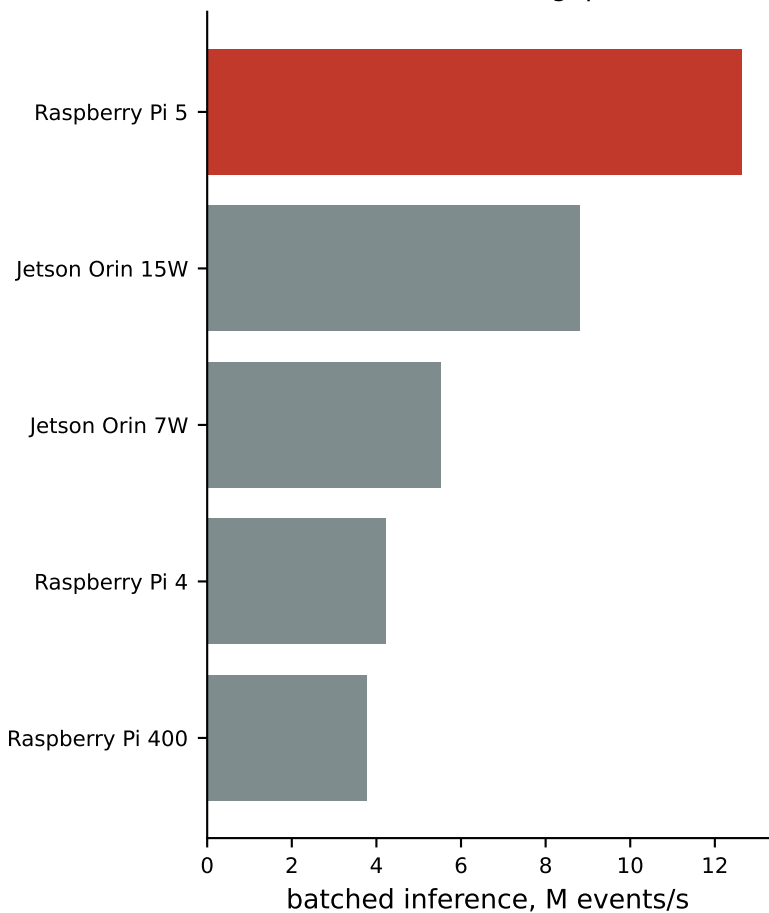
Device	steady s/round	6-round s/round	img/s	peak RSS MiB
Pi 5 (CPU)	0.057	0.234	10,274	326
Jetson 15W (CPU)	0.065	0.438	5,486	757
Jetson 15W (CUDA)	0.089	0.481	4,990	1097
Pi 4 (CPU)	0.162	0.900	2,665	323

Table C — Event gateway, as-fast-as-possible (200k synthetic events)

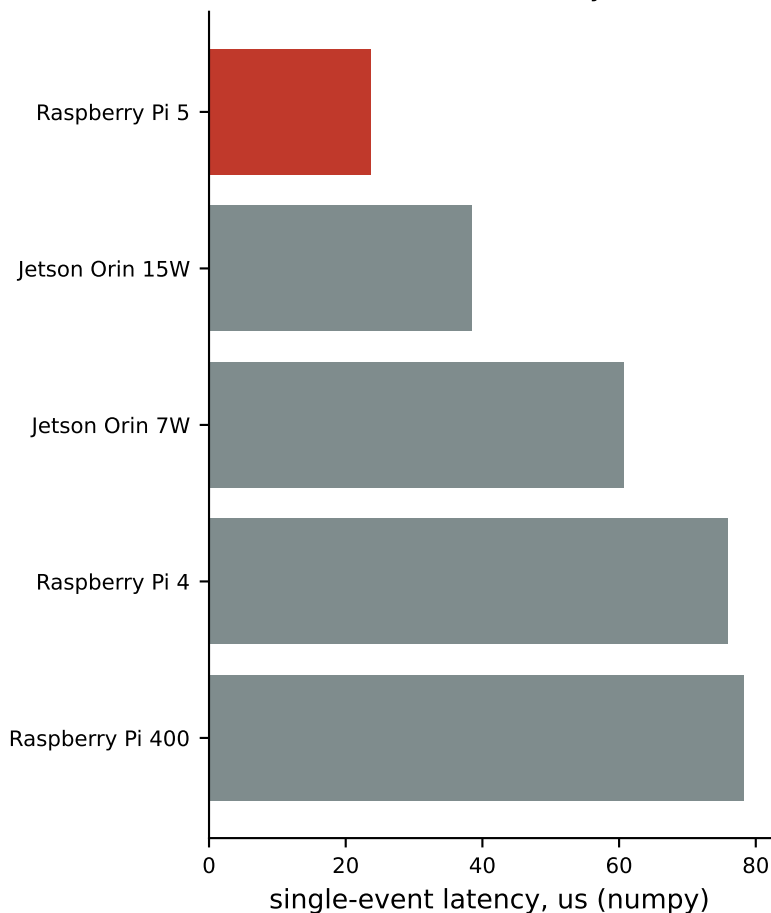
policy	Pi 4 us/ev	Pi 5 us/ev	speedup
coincidence	18.0	4.0	4.5x
adc	26.1	6.8	3.9x
mlp	74.0	19.3	3.8x
hybrid	75.9	20.1	3.8x

Pi 5 update — cross-device charts

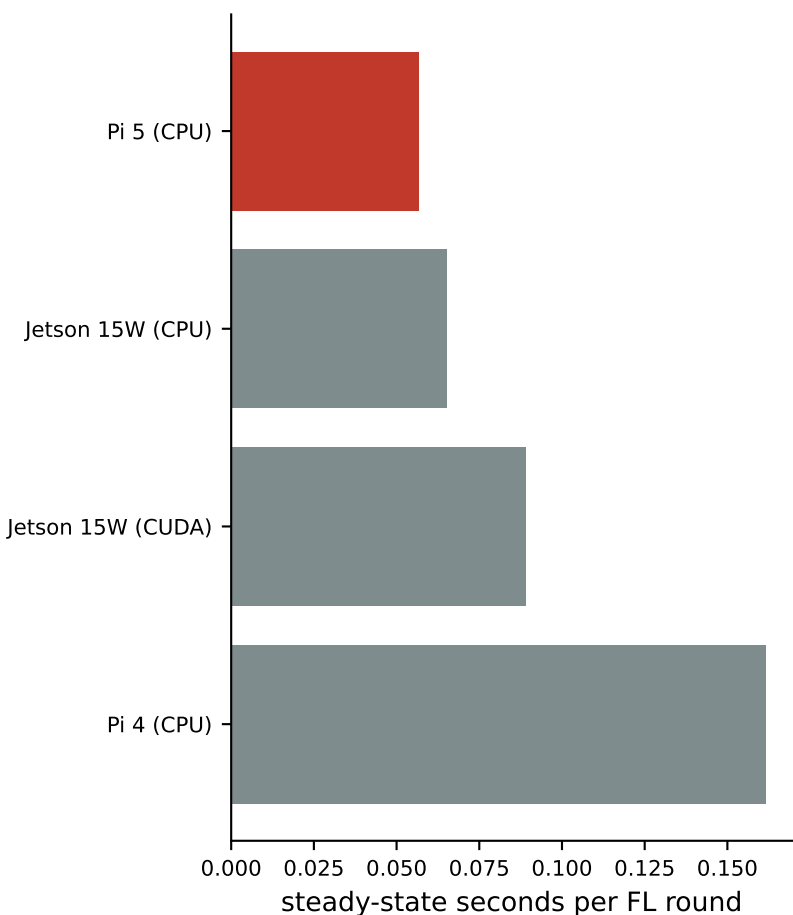
Inference throughput



Inference latency



FL client round time



Event gateway per policy

